

TIC-TAC-TOE Algorithm

2-Player · 3×3 Grid · C++ Implementation

X | O

1 Initialization

BOARD	Fill all 9 cells of the 3×3 grid with empty spaces.
PLAYERS	Player 1 → X Player 2 → O
TURN	Set current player to X (Player 1 goes first).
STATE	Set gameOver flag to false.

2 Main Game Loop

1	Display Board	Print the current state of the 3×3 grid.
2	Get Input	Prompt the current player for row & column (1–3).
3	Validate & Place	Convert to 0-based. Reject if out-of-range or occupied.
4	Check Win	Scan rows, columns & diagonals for 3-in-a-row.
5	Check Draw	If all 9 cells filled and no winner → draw.
6	Switch Player	Toggle: X → O or O → X. Repeat loop.

3 Win Detection (checkWin)

Return true if all three cells match the current player's mark in any of:

- **Rows:** `board[i][0] == board[i][1] == board[i][2]` for $i \in \{0,1,2\}$
- **Columns:** `board[0][j] == board[1][j] == board[2][j]` for $j \in \{0,1,2\}$
- **Main Diagonal:** `(0,0) → (1,1) → (2,2)`
- **Anti-Diagonal:** `(0,2) → (1,1) → (2,0)`

4 Control Flow Summary

